Worcester Polytechnic Institute

DS 504 Big Data Analytics

05/06/2025

TOPIC MODELING COVID-19 RESEARCH WITH AI AND NLP

A Comparative Study using TF-IDF, BERT, LDA, NMF, and BERTopic

OUR TEAM

Dinesh Kodwani

Guruganesh Holla

Jose Fabrizio Filizzola

Kathryn Butziger

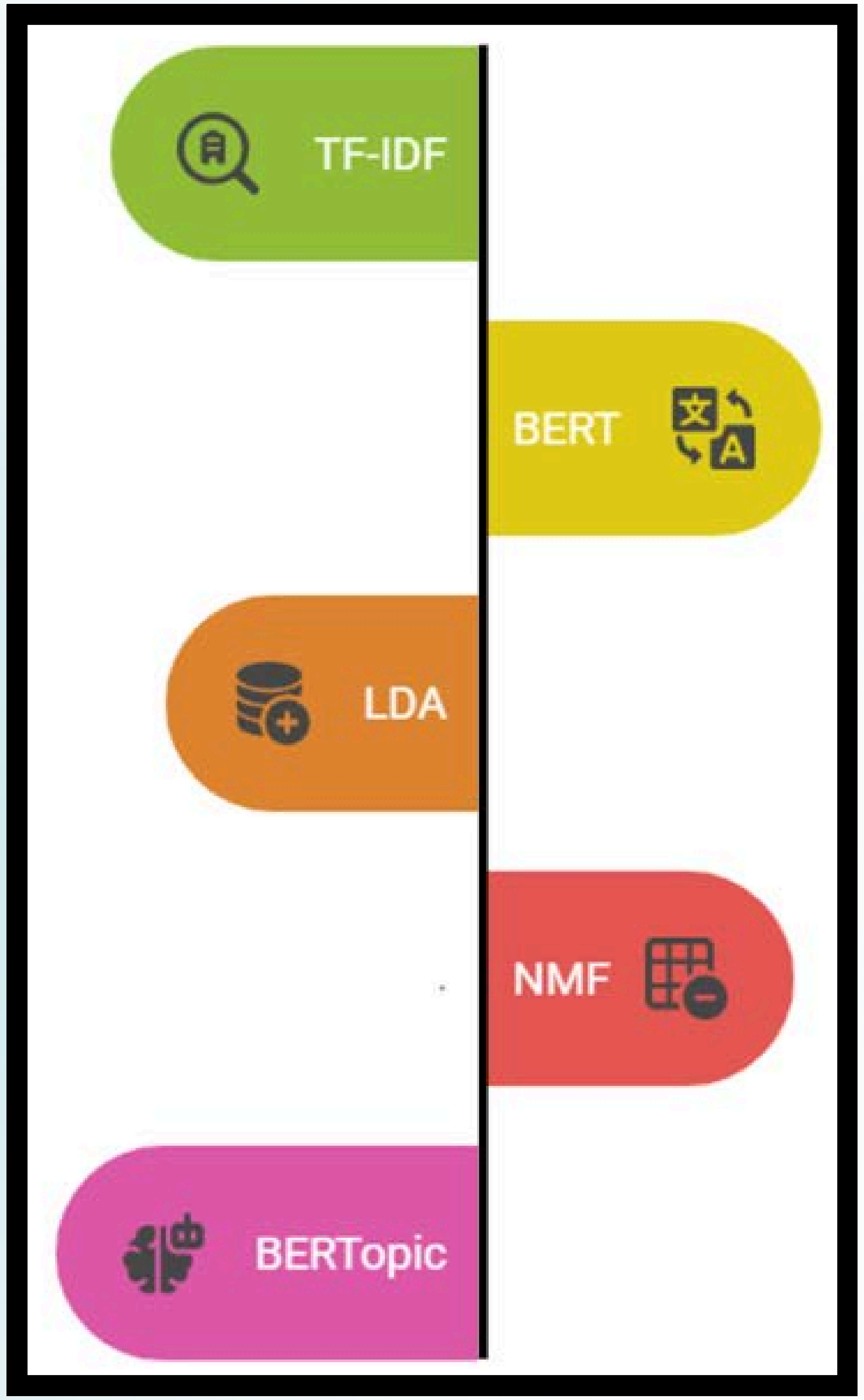
Mandar Nadkarni

Parth Surve

Steffi Dorothy

PROJECT PREVIEW

- COVID-19 research explosion → overwhelming literature
- **Goal** : Use AI + NLP to help researchers extract insights
- **Approach** : Combine TF-IDF, BERT, K-Means, LDA, NMF, BERTopic



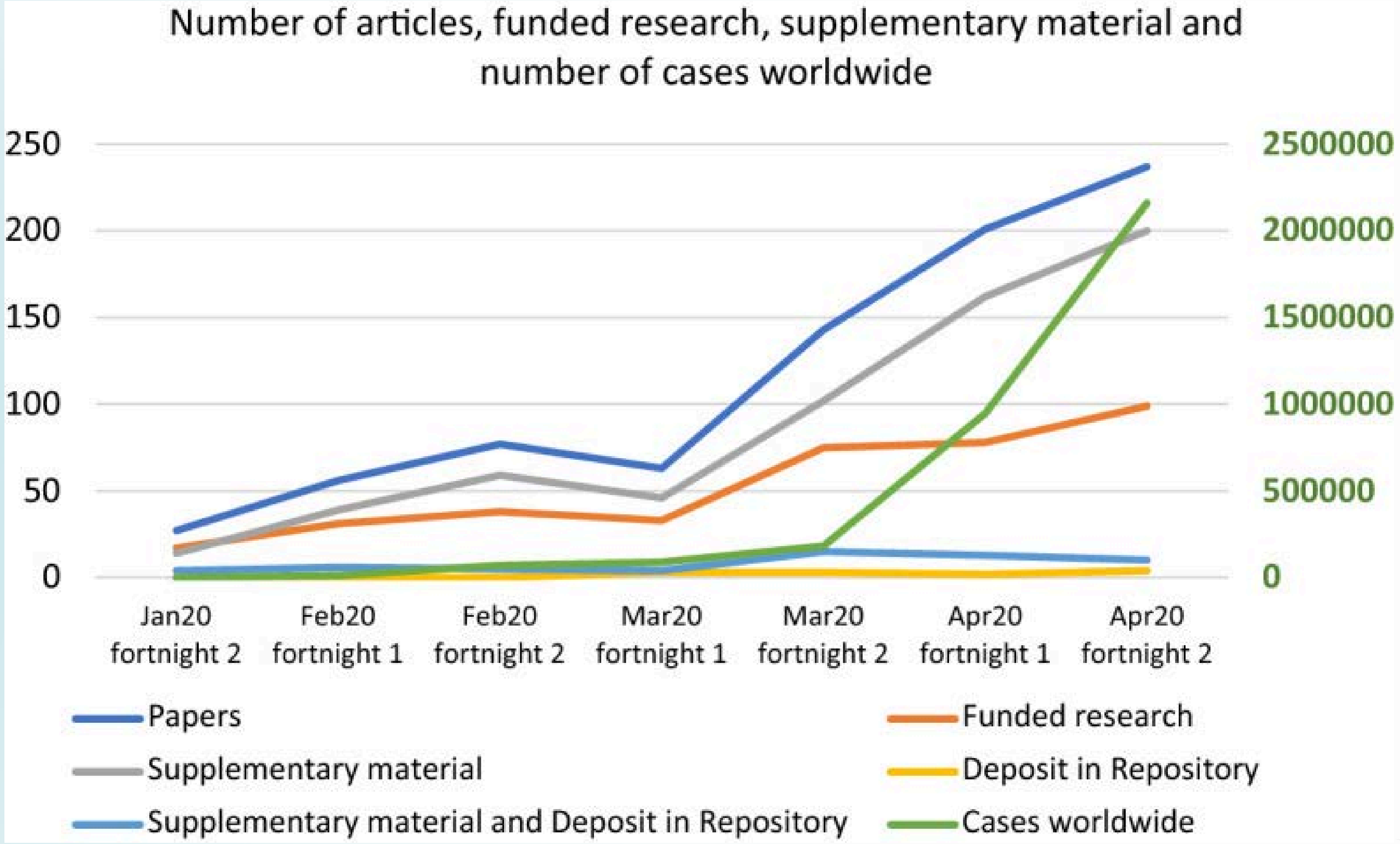
BACKGROUND

COVID-19 Research Surge : The pandemic led to an overwhelming volume of research papers.

Relevance : Extracting insights from this vast body of research is crucial for public health.

NLP : Topic modeling techniques can automate analysis and summarize findings.

Our Goal : Use BERT, NMF, and K-Means to categorize topics from COVID-19 papers.



OBJECTIVES

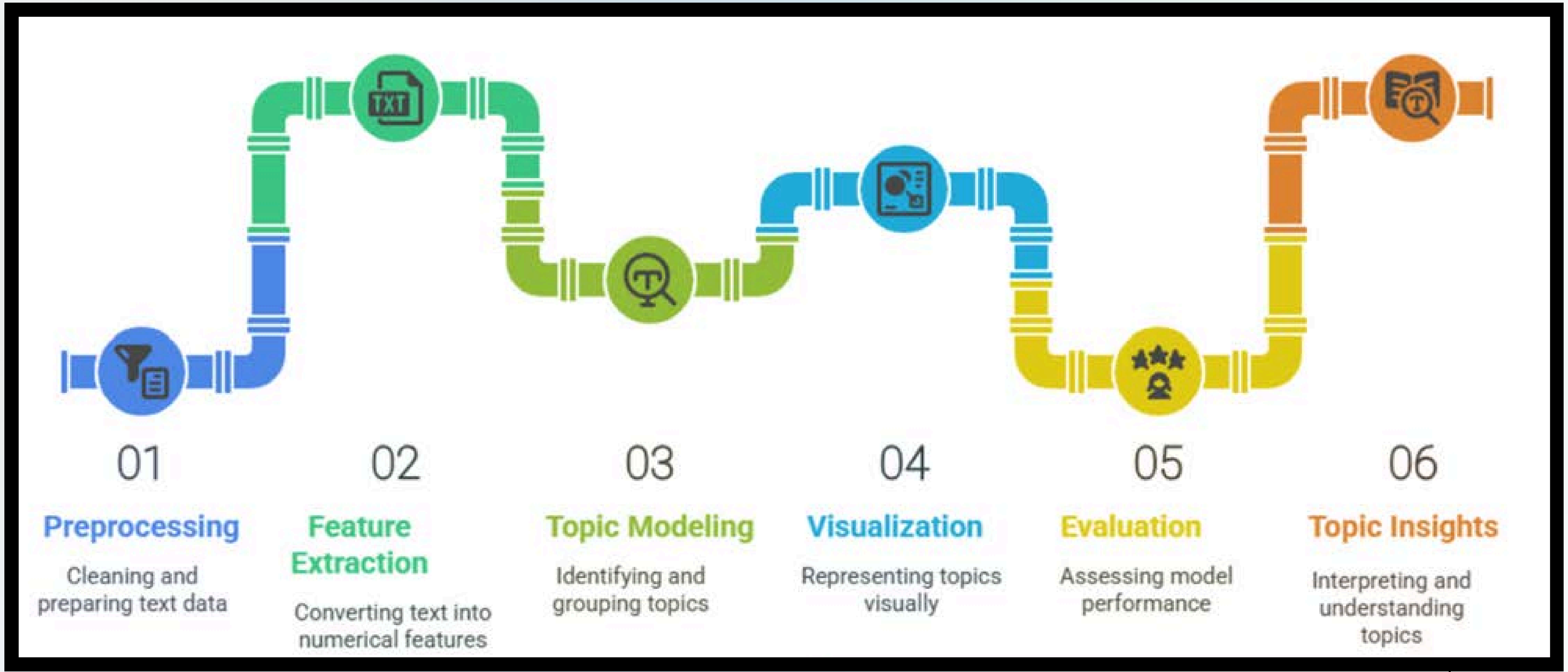
TOPIC MODELING

Use AI-driven techniques like LDA, BERT, NMF, and K-Means clustering to categorize and extract meaningful topics from a large set of COVID-19 research papers.

AUTOMATED INSIGHT EXTRACTION

Efficiently summarize and uncover key themes in COVID-19 research to help researchers and policymakers identify trends and make informed decisions.

METHODOLOGY



METHODOLOGY

Data Collection

Preprocessing

Feature Extraction

Modeling Techniques:

TF-IDF + K-Means clustering

BERT embeddings + K-Means clustering

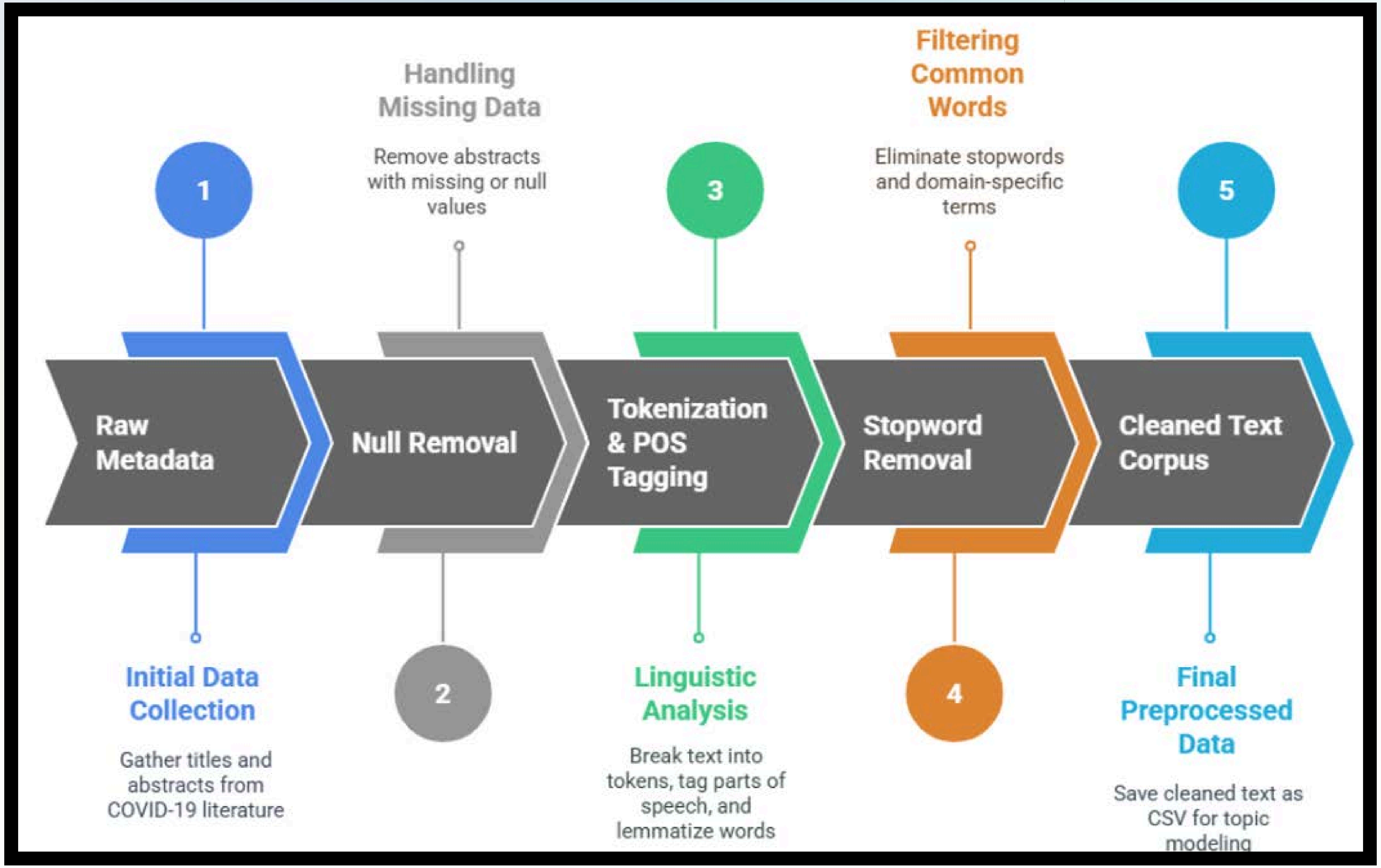
LDA (Latent Dirichlet Allocation)

NMF (Non-negative Matrix Factorization)

BERTopic

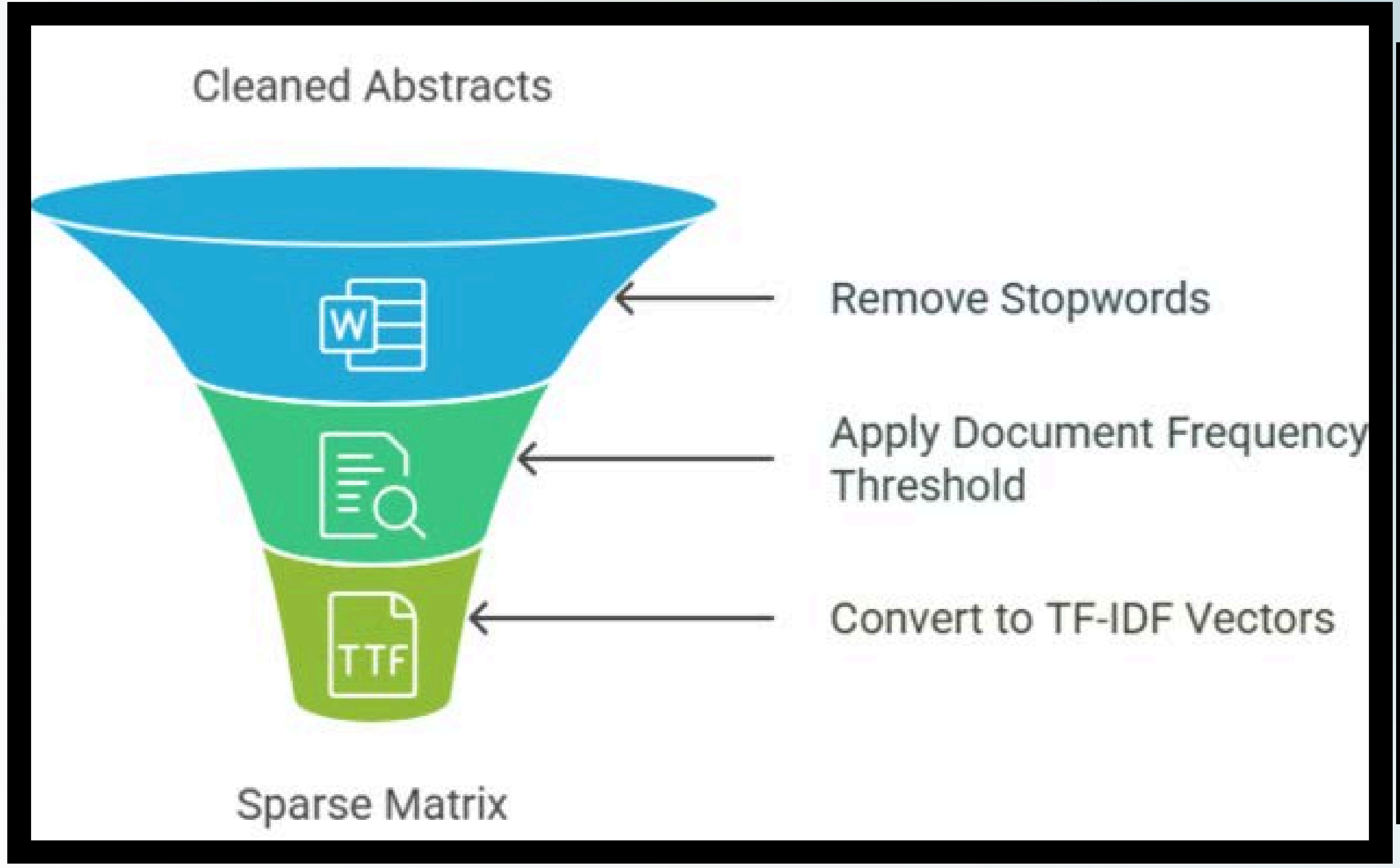
Demo app comparing LDA, NMF, BERTopic on real data

PREPROCESSING

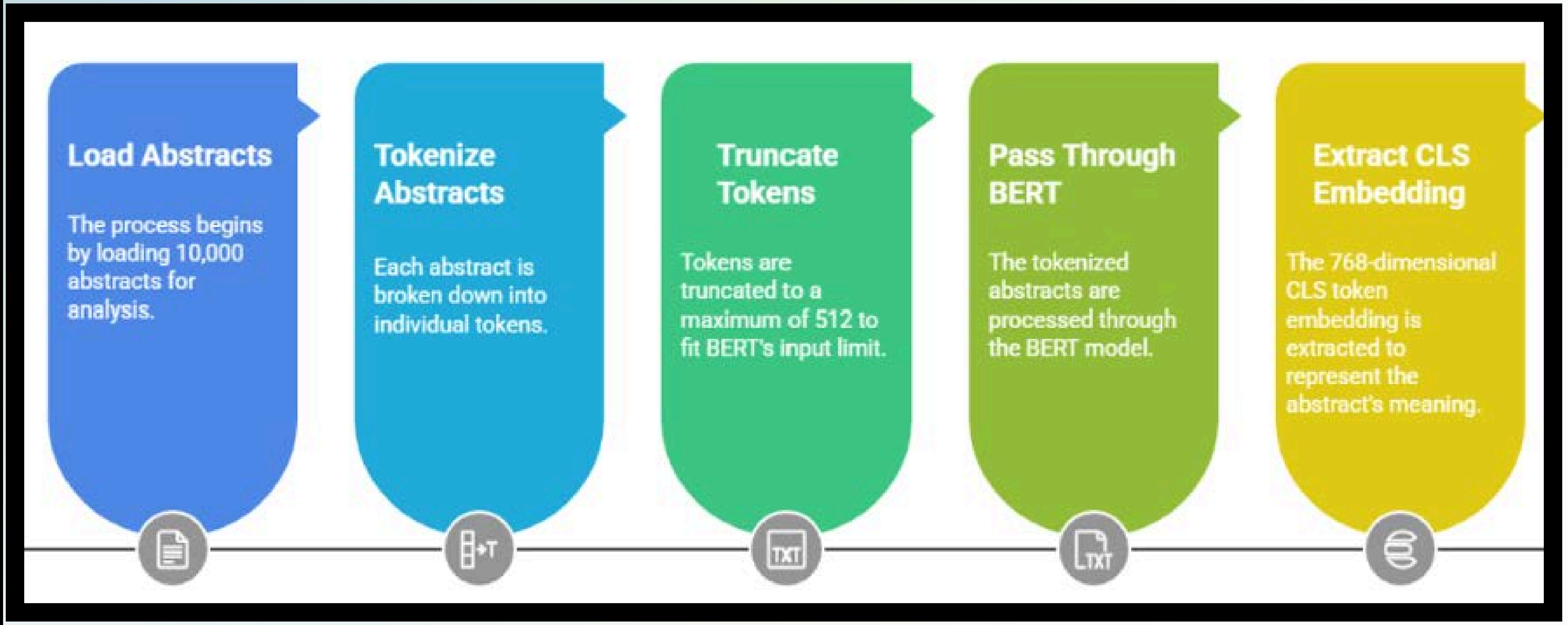


FEATURE EXTRACTION

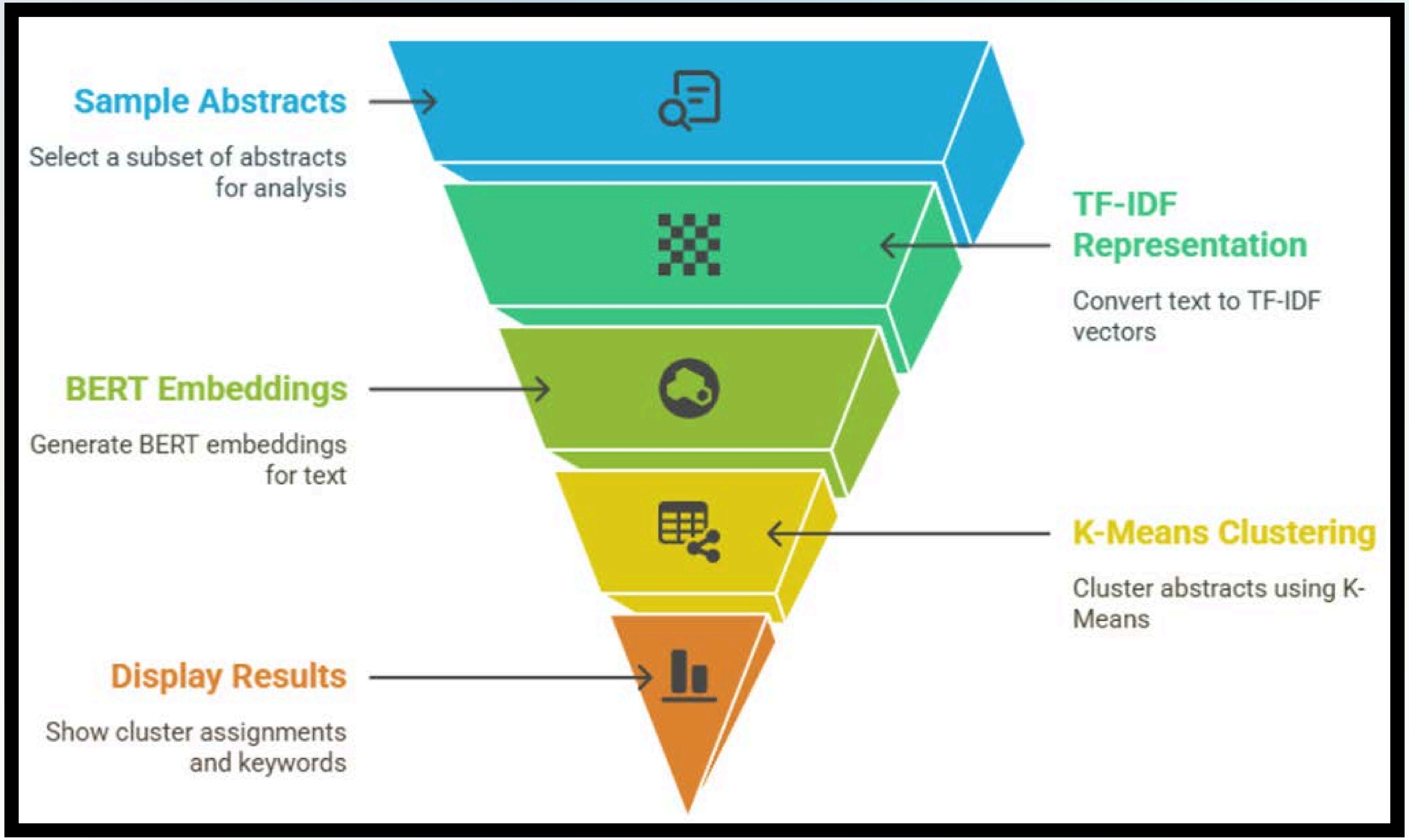
TF-IDF Feature Extraction



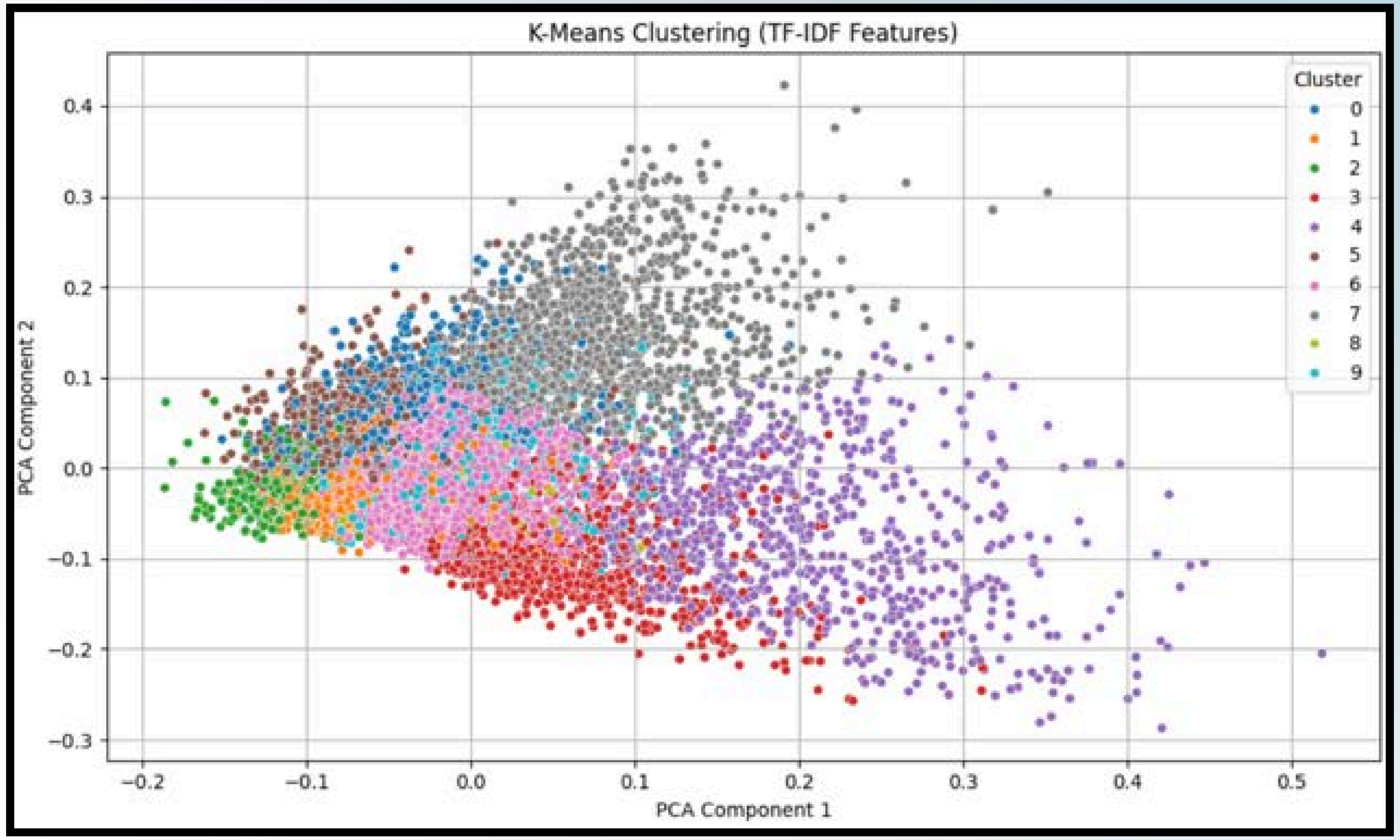
BERT Embedding Generation

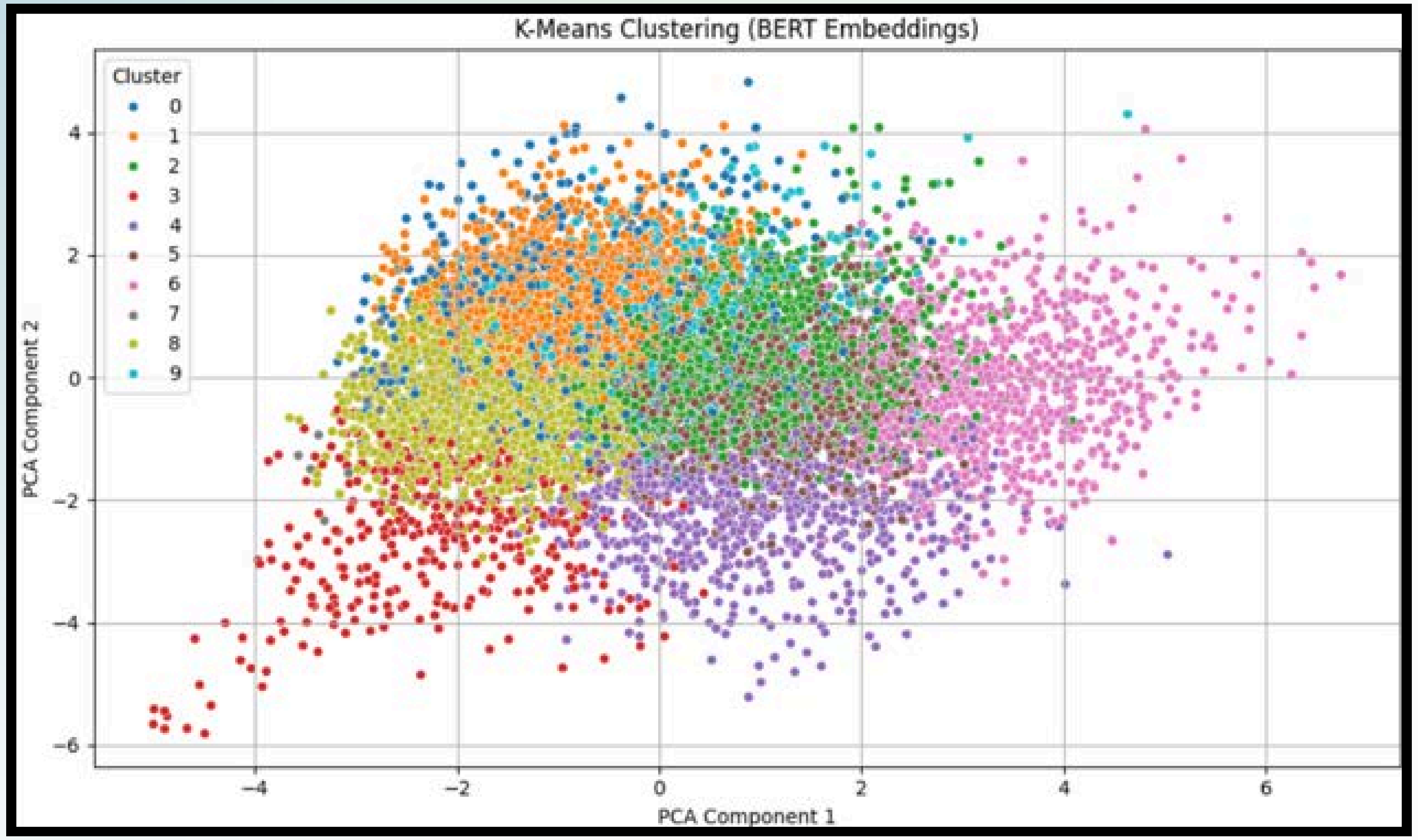


TOPIC MODELING WITH TF-IDF AND BERT EMBEDDINGS

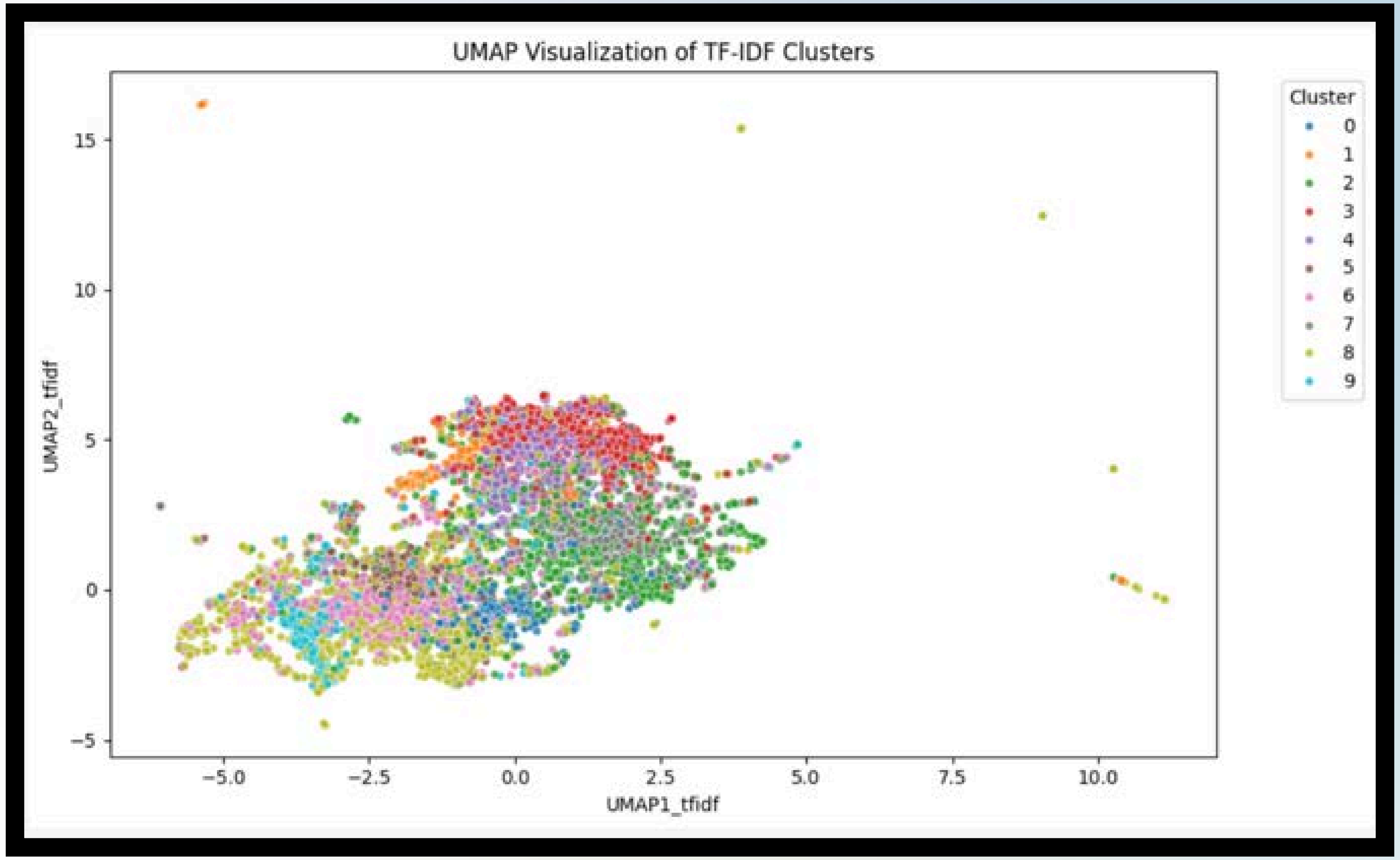


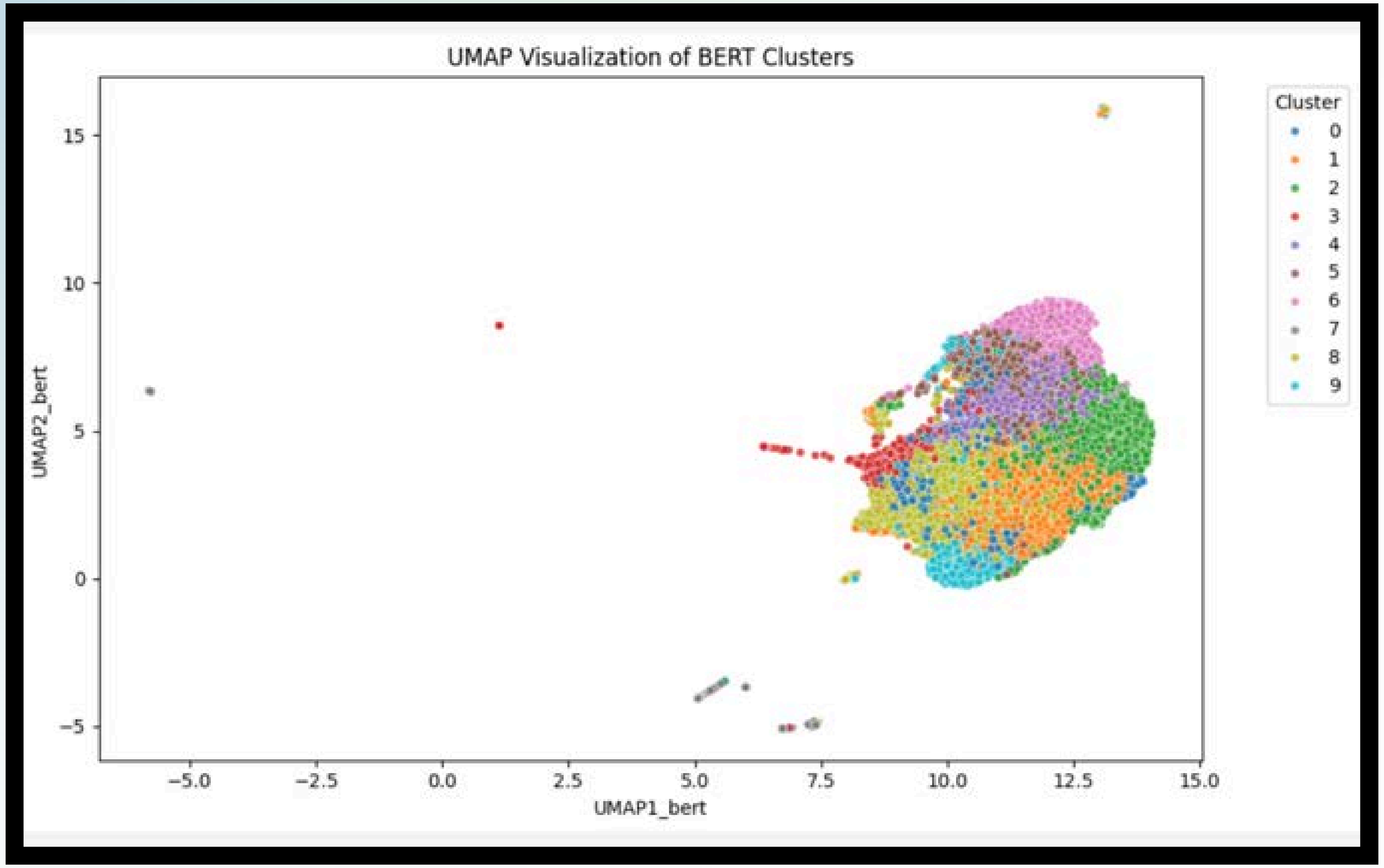
VISUALIZING DOCUMENT CLUSTERS WITH PCA





UMAP-BASED TOPIC VISUALIZATION





KEY TECHNIQUES: LDA, NMF, BERT

01

LDA (Latent Dirichlet Allocation)

- A probabilistic topic model.
- Example : In COVID papers, LDA might identify topics like "vaccines," "lockdowns," or "treatment trials."

02

NMF (Non-negative Matrix Factorization)

- A matrix factorization method for topic modeling.
- Compared to LDA, it often gives more interpretable, sparse topics.

03

BERT (Bidirectional Encoder Representations from Transformers)

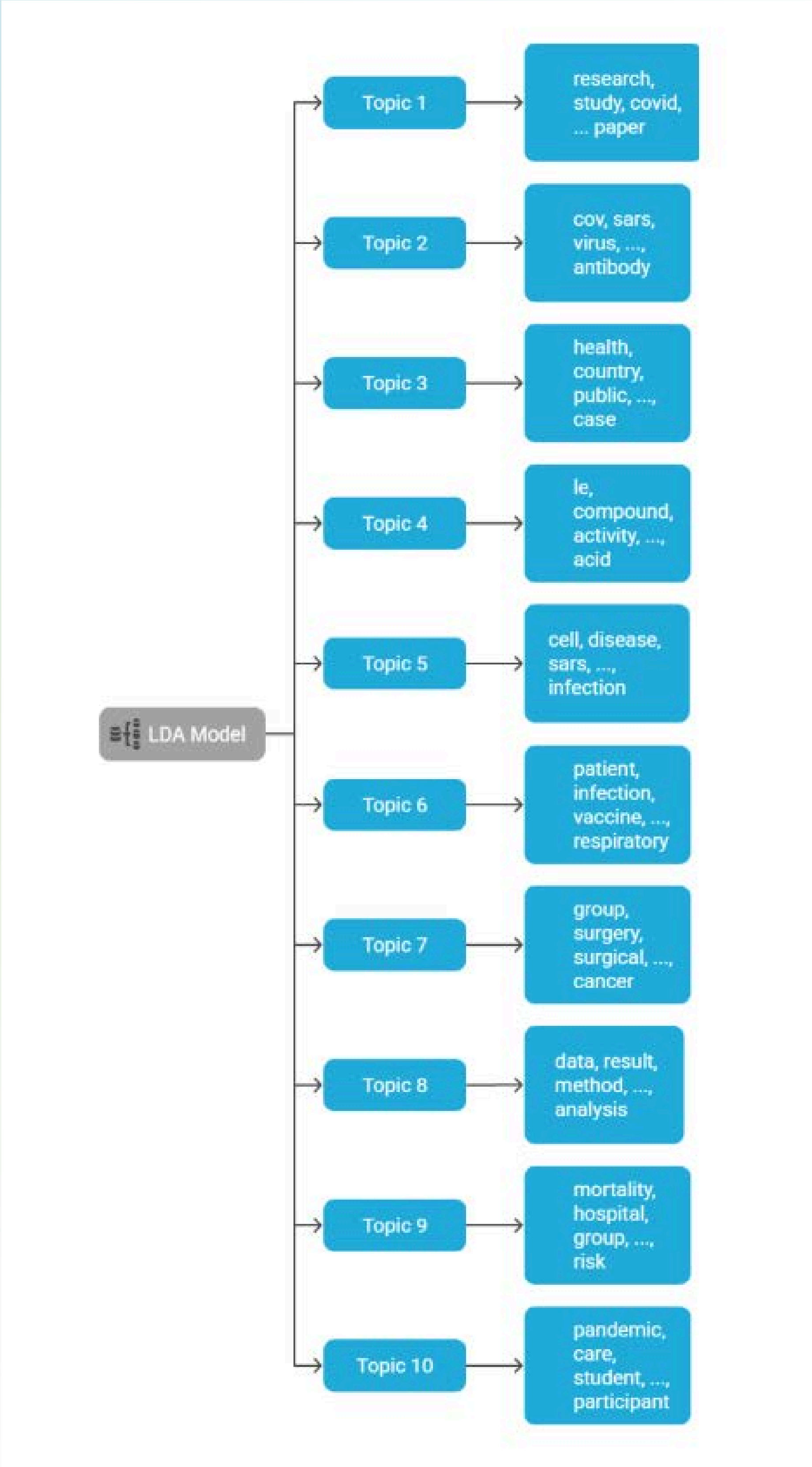
- Unlike LDA and NMF, BERT captures the context and meaning of words using word embeddings.

TOPIC DISCOVERY USING LATENT DIRICHLET ALLOCATION (LDA)

01

LDA (Latent Dirichlet Allocation)

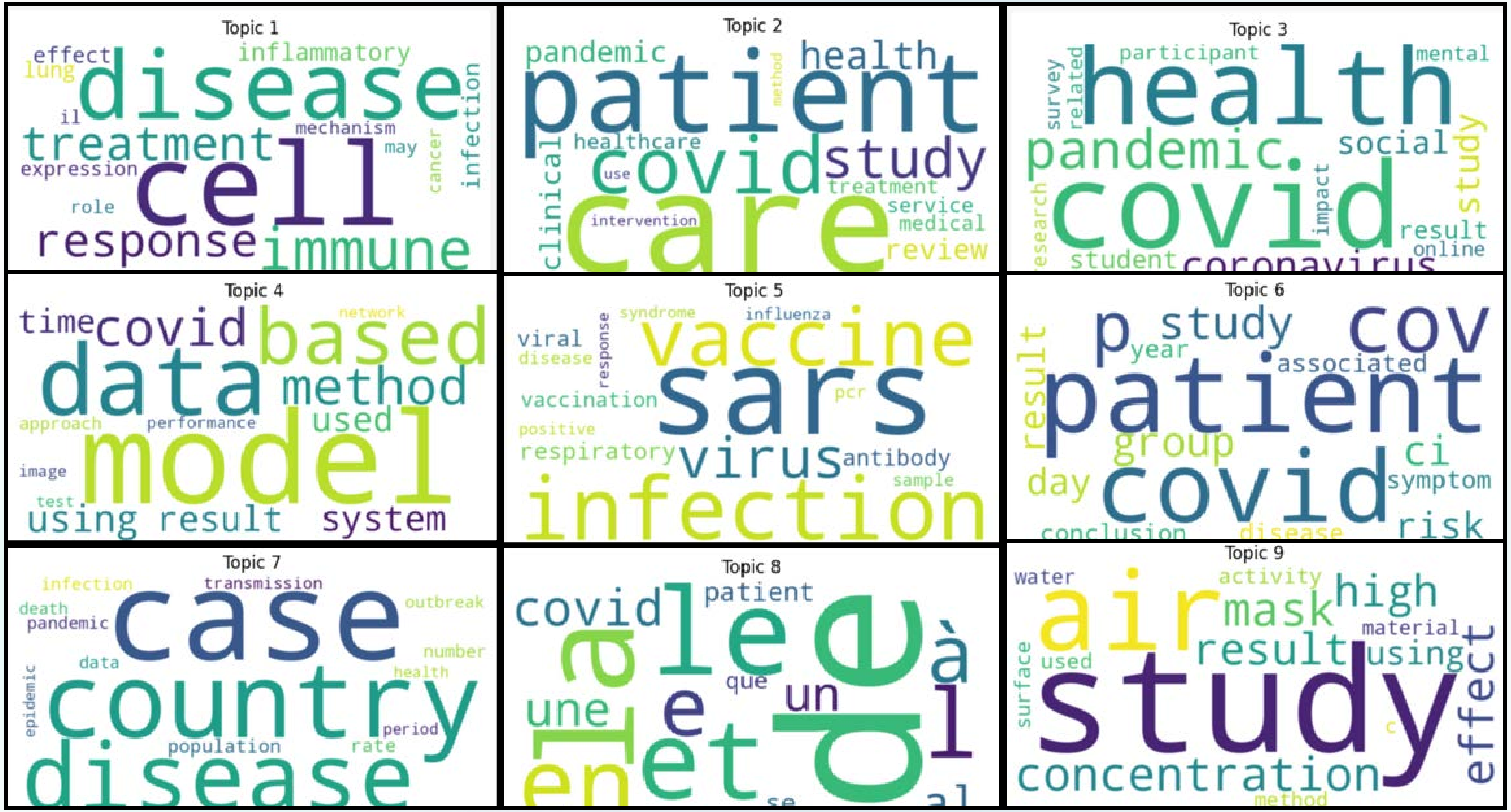
- **Approach :** LDA using Gensim on bag-of-words corpus
- **Parameters :** 10 topics, 10 passes, random state = 42



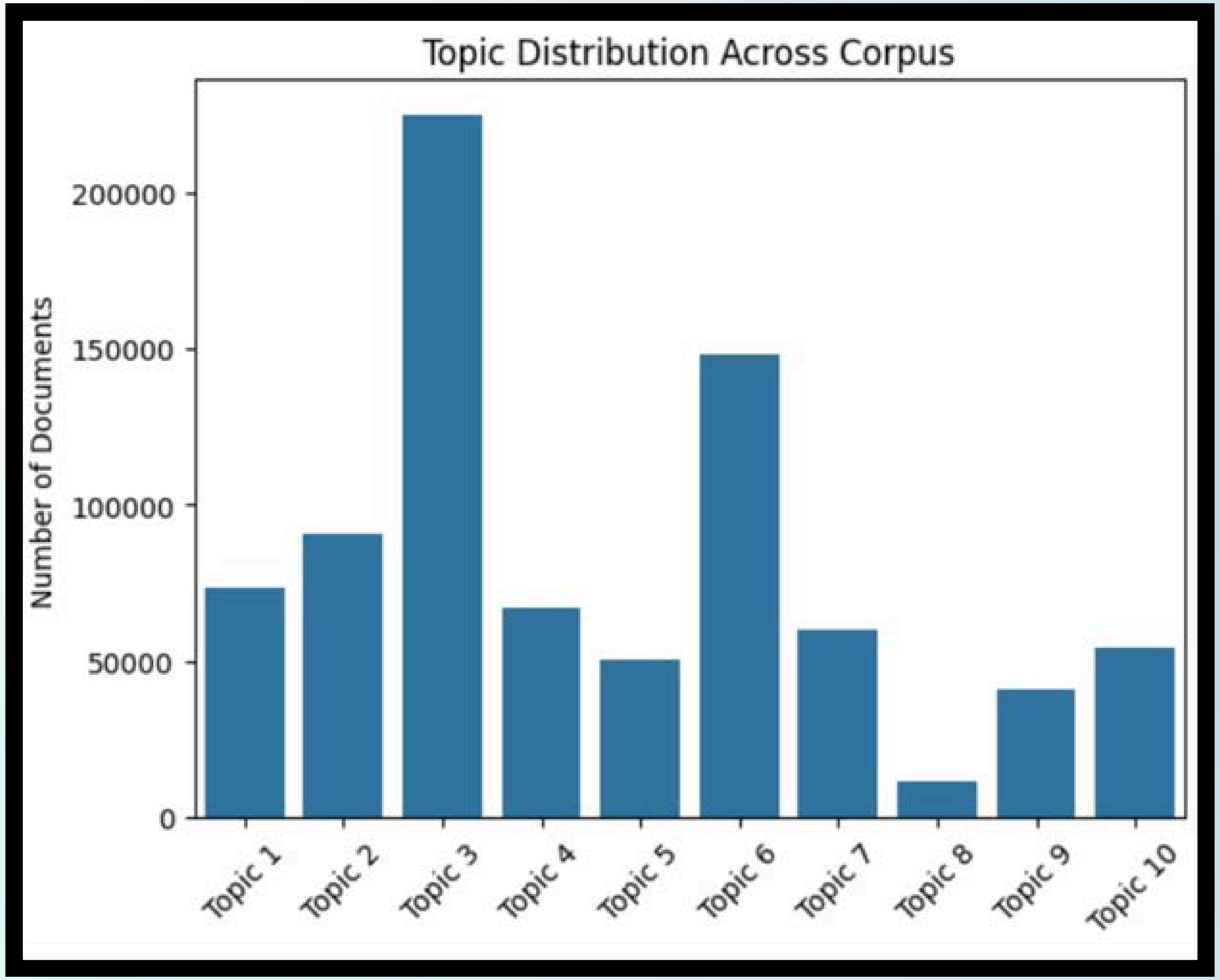
RESULT 1 - TOP KEYWORDS PER TOPIC

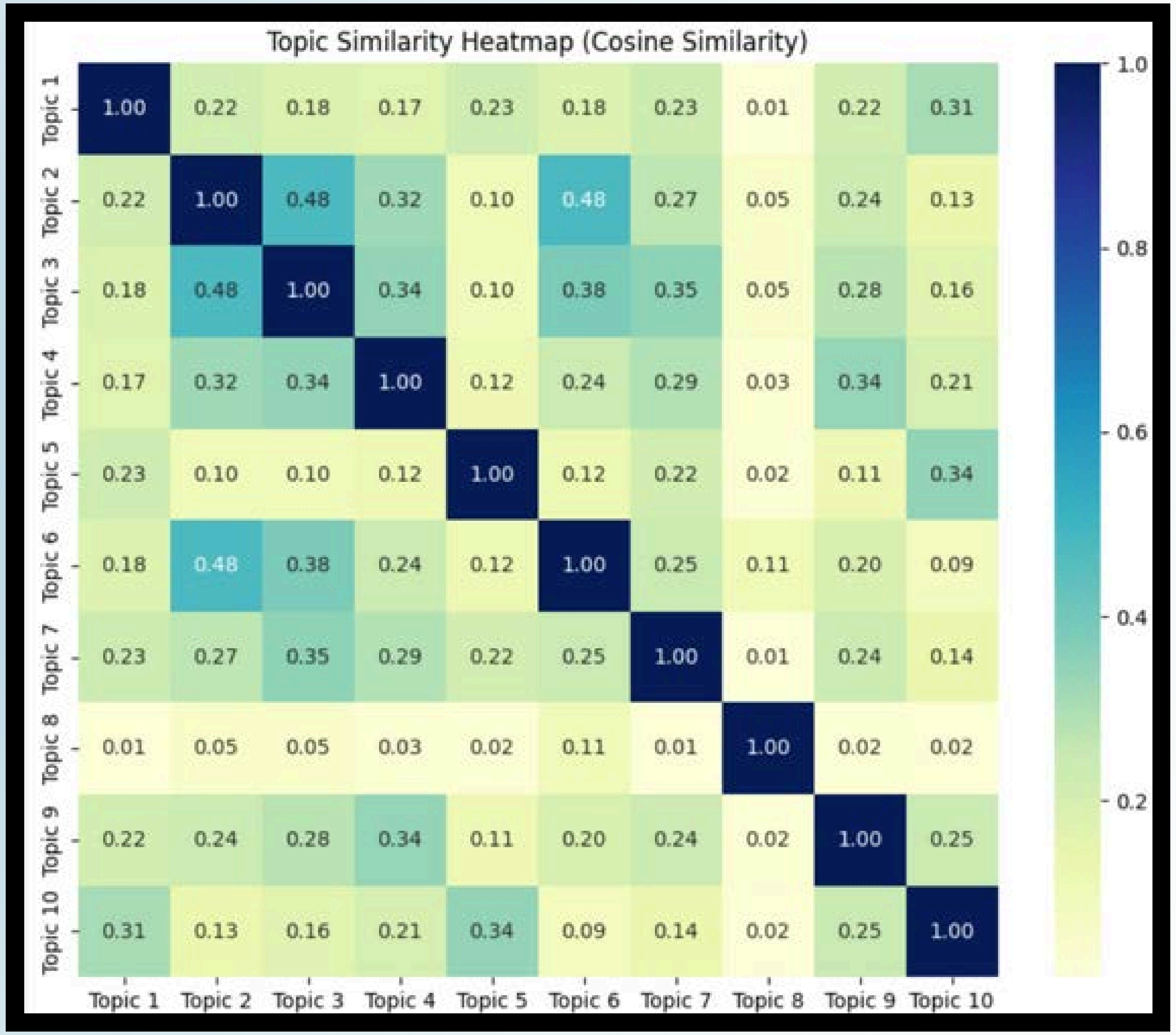
```
Topic 1: 0.023*"cell" + 0.015*"disease" + 0.009*"immune" + 0.008*"treatment" + 0.008*"response" + 0.007*"inflammatory" + 0.007*"infection" + 0.006*"lung" + 0.006*"expression" + 0.006

Topic 2: 0.020*"care" + 0.017*"patient" + 0.011*"covid" + 0.010*"study" + 0.009*"health" + 0.008*"pandemic" + 0.007*"review" + 0.007*"clinical" + 0.007*"healthcare" + 0.007*"service"

Topic 3: 0.020*"covid" + 0.016*"health" + 0.013*"pandemic" + 0.013*"coronavirus" + 0.010*"study" + 0.008*"social" + 0.005*"student" + 0.005*"result" + 0.005*"participant" + 0.005*"im

Topic 4: 0.021*"model" + 0.011*"data" + 0.011*"based" + 0.011*"method" + 0.011*"covid" + 0.008*"using" + 0.007*"system" + 0.007*"result" + 0.007*"used" + 0.006*"time"

Topic 5: 0.049*"sars" + 0.030*"infection" + 0.028*"vaccine" + 0.024*"virus" + 0.018*"respiratory" + 0.016*"antibody" + 0.015*"viral" + 0.014*"vaccination" + 0.010*"disease" + 0.010*"

Topic 6: 0.047*"patient" + 0.039*"covid" + 0.027*"cov" + 0.019*"p" + 0.012*"study" + 0.010*"group" + 0.010*"ci" + 0.008*"risk" + 0.008*"day" + 0.008*"result"

Topic 7: 0.018*"case" + 0.012*"country" + 0.011*"disease" + 0.011*"rate" + 0.011*"population" + 0.010*"number" + 0.009*"data" + 0.009*"pandemic" + 0.009*"infection" + 0.008*"death"

Topic 8: 0.086*"de" + 0.031*"la" + 0.025*"le" + 0.021*"et" + 0.017*"en" + 0.016*"l" + 0.012*"e" + 0.011*"à" + 0.010*"covid" + 0.009*"une"

Topic 9: 0.007*"study" + 0.006*"air" + 0.006*"mask" + 0.006*"concentration" + 0.006*"effect" + 0.005*"high" + 0.005*"result" + 0.005*"using" + 0.005*"activity" + 0.005*"used"

Topic 10: 0.023*"protein" + 0.014*"sars" + 0.014*"virus" + 0.012*"gene" + 0.010*"human" + 0.010*"cell" + 0.009*"viral" + 0.009*"binding" + 0.008*"host" + 0.008*"rna"
```

RESULT II - VISUALIZING TOPICS WITH WORD CLOUDS

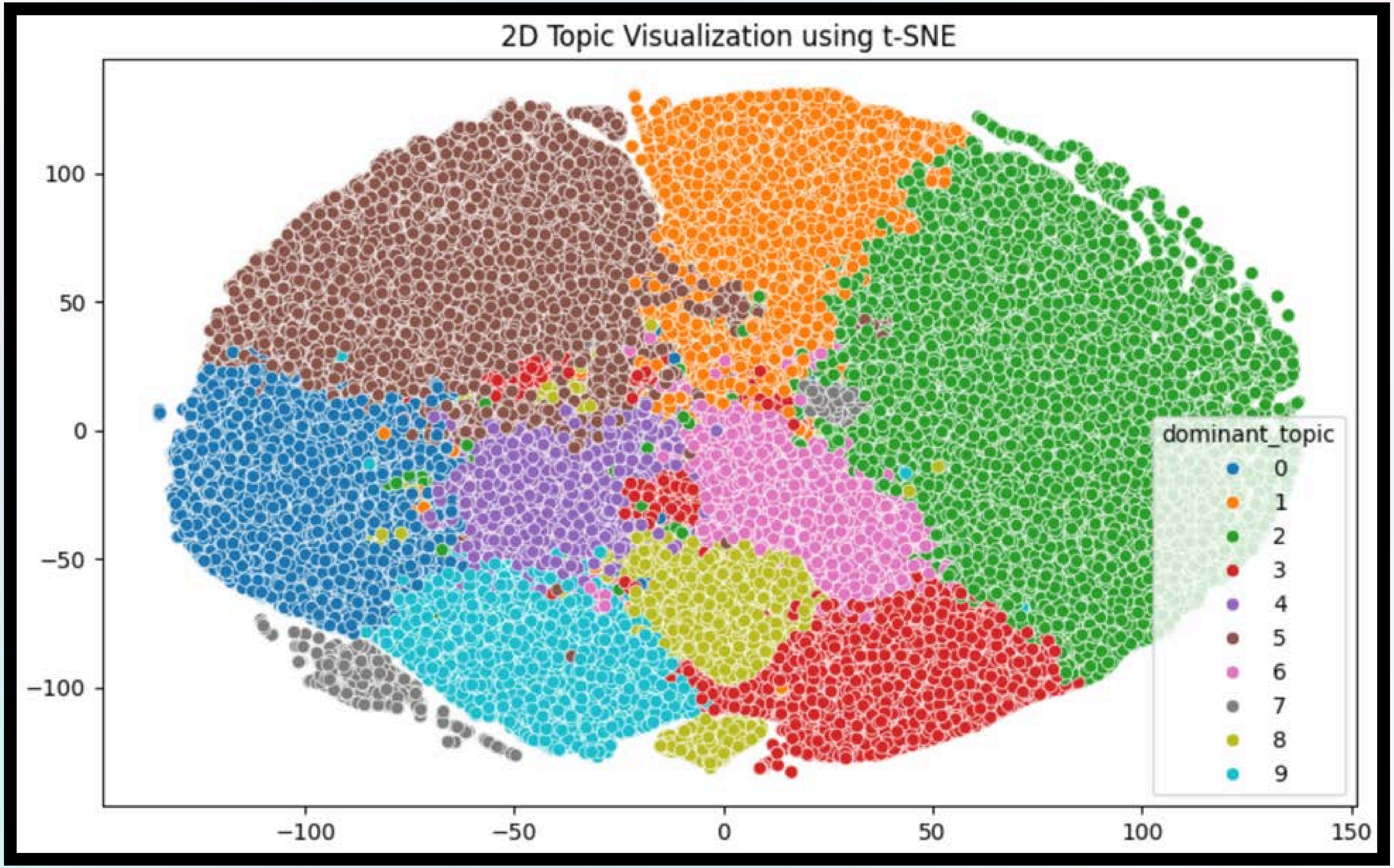


RESULT III - TOPIC DISTRIBUTION ACROSS CORPUS





RESULT IV



2D Topic Visualization using t-SNE
100
50
0
−50
−100
−100
−50
0
50
100
150
dominant_topic
0
1
2
3
4
5
6
7
8
9

TOPIC DISCOVERY USING NMF

02

NMF on TF-IDF matrix

Parameters :

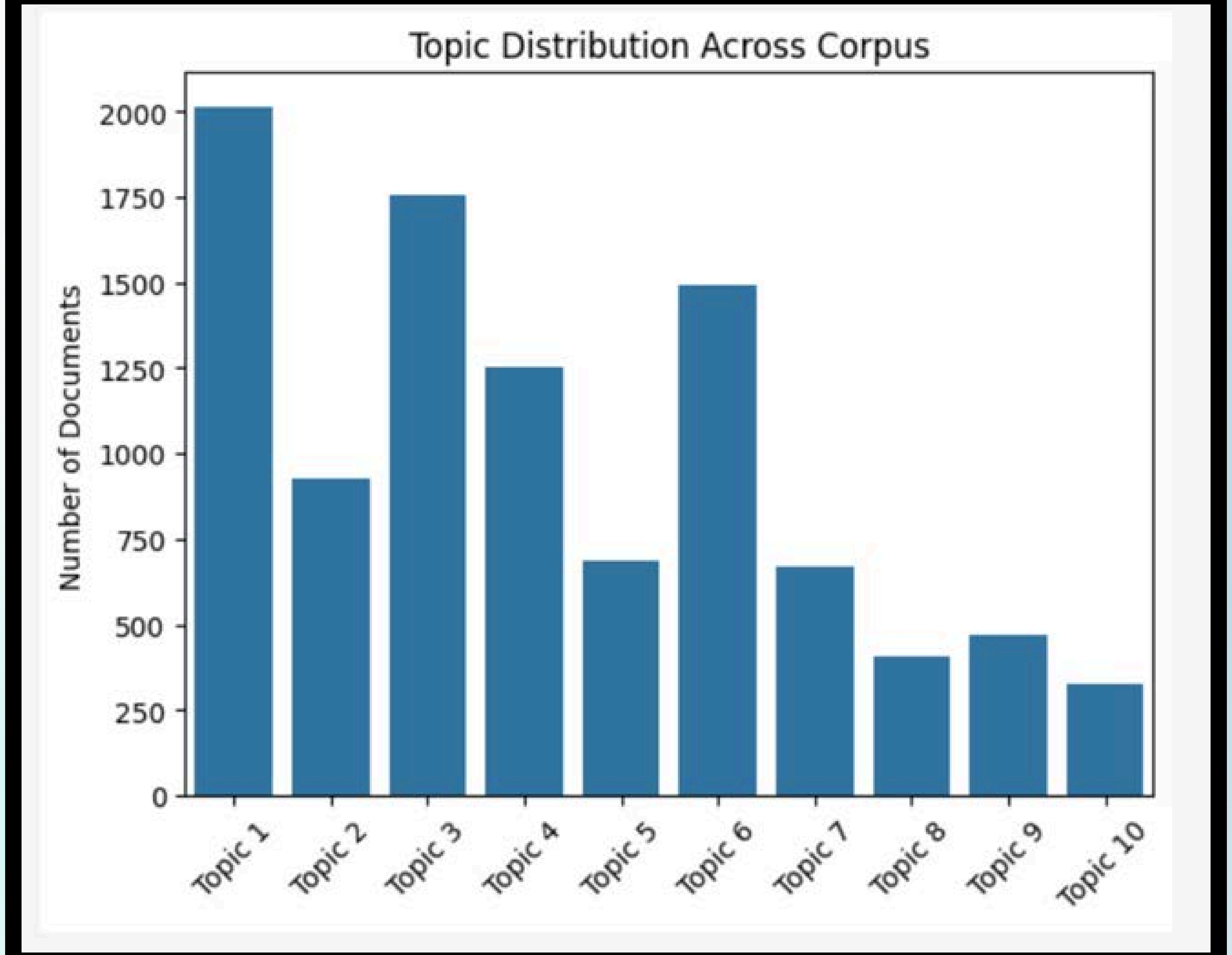
- 10 topics (n_components = 10)
- Stop words removed (English)
- max_df = 0.95

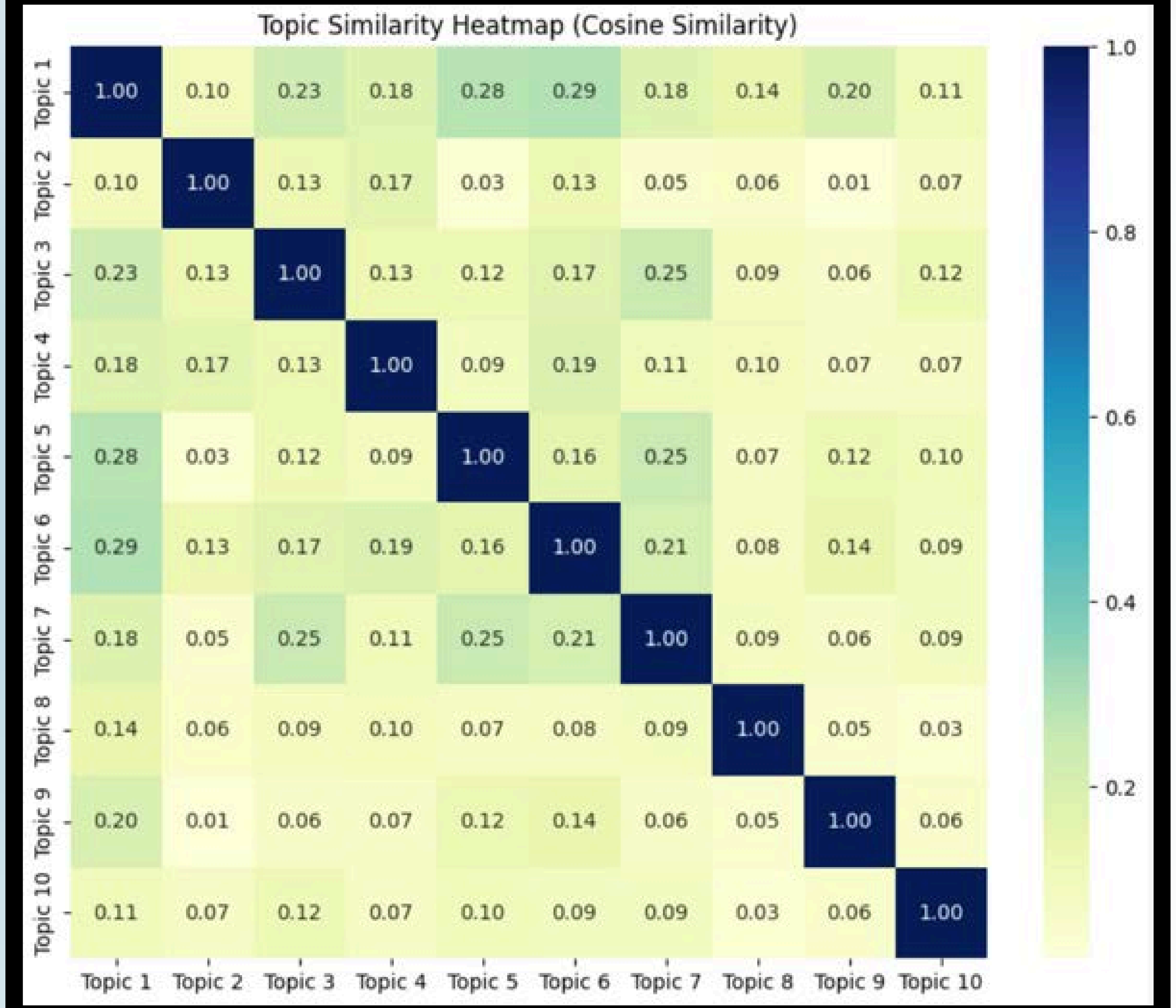
Sklearn implementation of NMF w/ 10,000 rows sampled from our cleaned dataset

RESULT 1 - TOP KEYWORDS PER TOPIC

```
Topic 1: impact,healthcare,service,research,public,social,care,pandemic,covid,health
Topic 2: antibody,acute,severe,virus,syndrome,coronavirus,respiratory,infection,cov,sars
Topic 3: cancer,symptom,care,severe,treatment,hospital,clinical,disease,covid,patient
Topic 4: human,host,expression,gene,viral,immune,drug,virus,protein,cell
Topic 5: physical,study,symptom,participant,stress,psychological,health,depression,anxiety,mental
Topic 6: proposed,test,method,time,based,number,epidemic,data,case,model
Topic 7: factor,rate,associated,study,group,death,age,risk,mortality,ci
Topic 8: immunity,response,hesitancy,variant,vaccinated,dose,antibody,covid,vaccination,vaccine
Topic 9: virtual,medical,teacher,school,university,teaching,online,education,learning,student
Topic 10: symptom,infection,case,year,school,mi,pediatric,family,parent,child
```

RESULT II - TOPIC DISTRIBUTION ACROSS CORPUS





TOPIC DISCOVERY USING BERTOPIC

03

BERT embeddings + BERTopic

Parameters:

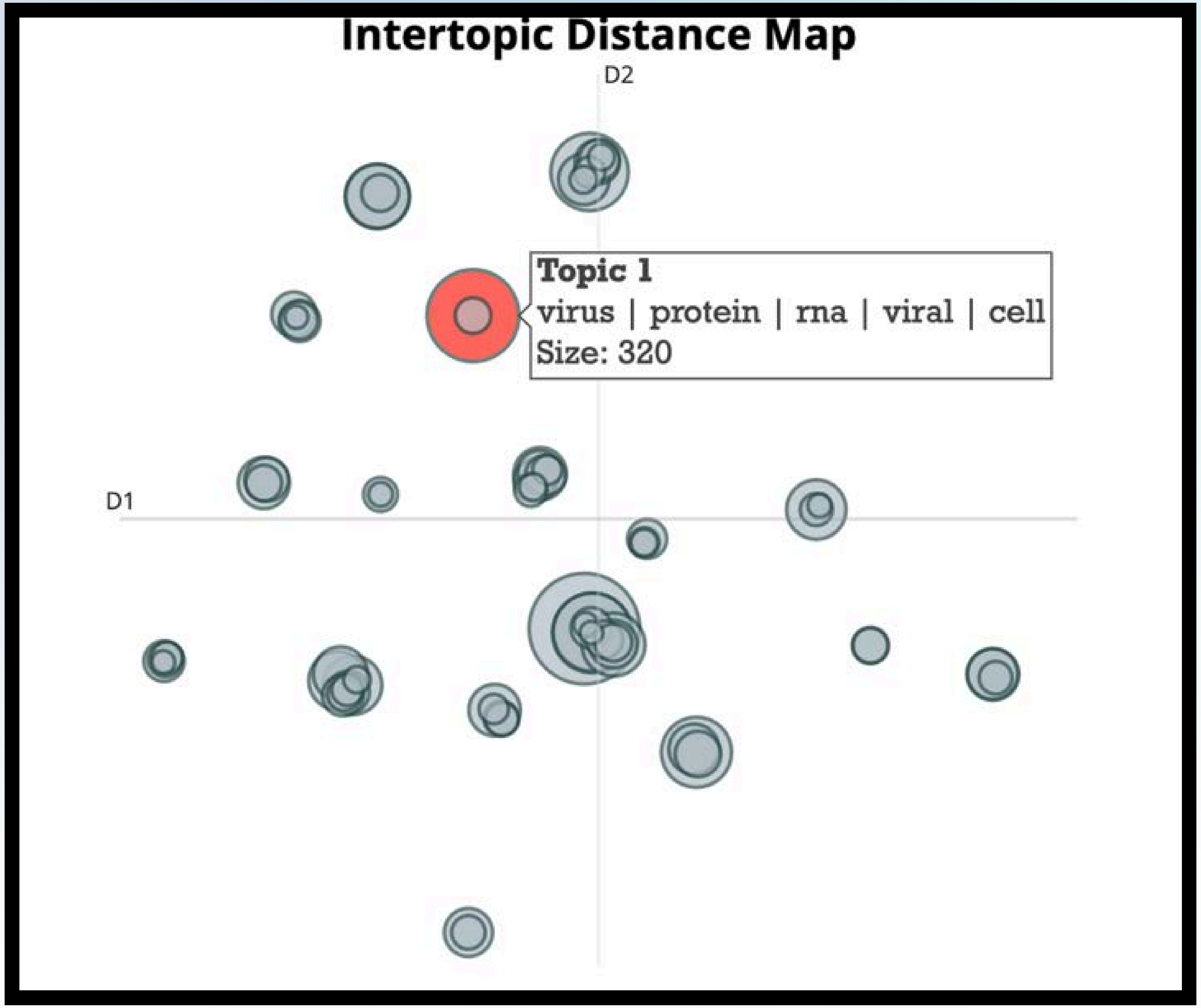
- Precomputed BERT embeddings (bert_embeddings_10k.npy)

CountVectorizer (stop words = English + custom, ngram 1–3, max_features = 5000)

UMAP dimensionality reduction (n_neighbors=15, n_components=5)

min_topic_size = 20

RESULTS BERTOPIC



GENERATED TOPICS

Topic 0: ['student', 'learning', 'education', 'teaching', 'online', 'teacher', 'educational', 'university', 'school', 'course']

Topic 1: ['virus', 'protein', 'rna', 'viral', 'cell', 'strain', 'genome', 'replication', 'sequence', 'infection']

Topic 2: ['care', 'dementia', 'service', 'caregiver', 'intervention', 'review', 'older', 'support', 'research', 'practice']

Topic 3: ['food', 'exercise', 'effect', 'obesity', 'physical', 'intervention', 'intake', 'participant', 'weight', 'nutrition']

Topic 4: ['cell', 'expression', 'mouse', 'inflammatory', 'induced', 'protein', 'il', 'inflammation', 'macrophage', 'ad']

Topic 5: ['anxiety', 'mental', 'depression', 'stress', 'psychological', 'suicide', 'fear', 'symptom', 'scale', 'social']

Topic 6: ['nanoparticles', 'surface', 'application', 'np', 'polymer', 'drug', 'adsorption', 'structure', 'nm', 'microfluidic']

Topic 7: ['sars', 'cov', 'protein', 'drug', 'binding', 'spike', 'coronavirus', 'protease', 'virus', 'mutation']

Topic 8: ['image', 'accuracy', 'algorithm', 'model', 'classification', 'deep', 'cnn', 'learning', 'neural', 'network']

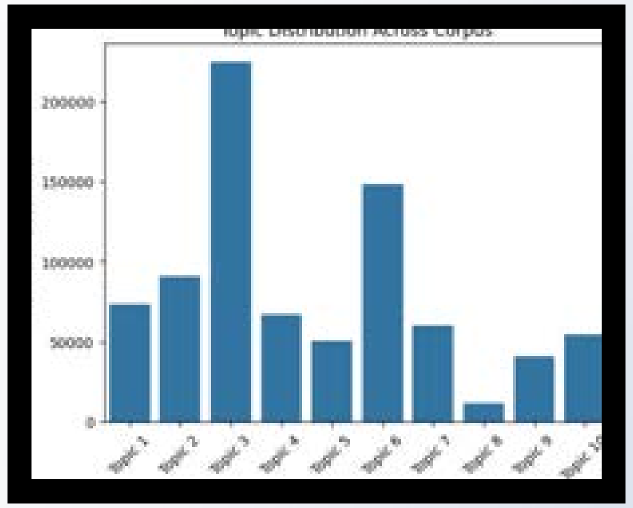
Topic 9: ['pain', 'joint', 'patient', 'knee', 'group', 'treatment', 'outcome', 'sedation', 'score', 'osteoarthritis']

INTERACTIVE TOPIC MODELING DEMO (LDA, NMF, BERTOPIC)

EVALUATION METRICS FOR TOPIC MODELING

MODEL	SILHOUETTE SCORE	COHERENCE SCORE
KMEANS (TF-IDF)	0.0038	N/A
KMEANS (BERT)	0.0361	N/A
LDA	N/A	0.570636
NMF	N/A	0.672544
BERTOPIC	0.554	0.738151

CONCLUSION



Successfully applied LDA, NMF, and BERTopic on ~10,000 documents

Extracted meaningful topics and top keywords from each method

BERTopic showed especially coherent and interpretable topics using BERT embeddings

THANK YOU

Presented by

Team 9

Q&A
SESSION